

32.1. Working Principle of a Transformer

A transformer is a static (or stationary) piece of apparatus by means of which electric power in one circuit is transformed into electric power of the same frequency in another circuit. It can raise or lower the voltage in a circuit but with a corresponding decrease or increase in current. The physical basis of a transformer is *mutual induction* between two circuits linked by a common magnetic flux. In its simplest form, it consists of two inductive coils which are electrically separated but magnetically linked through a path of low reluctance as shown in Fig. 32.1. The two coils possess high mutual inductance. If one coil is connected to a source of alternating voltage, an alternating flux is set up in the laminated core, most of which is linked with the other coil in which it produces mutually-induced e.m.f. (according to Faraday's Laws of Electromagnetic Induction $e = M dI/dt$). If the second coil circuit is closed, a current flows in it and so electric energy is transferred (entirely magnetically) from the first coil to the second coil. The first coil, in which electric energy is fed from the a.c. supply mains, is called *primary* winding and the other from which energy is drawn out, is called *secondary* winding. In brief, a transformer is a device that

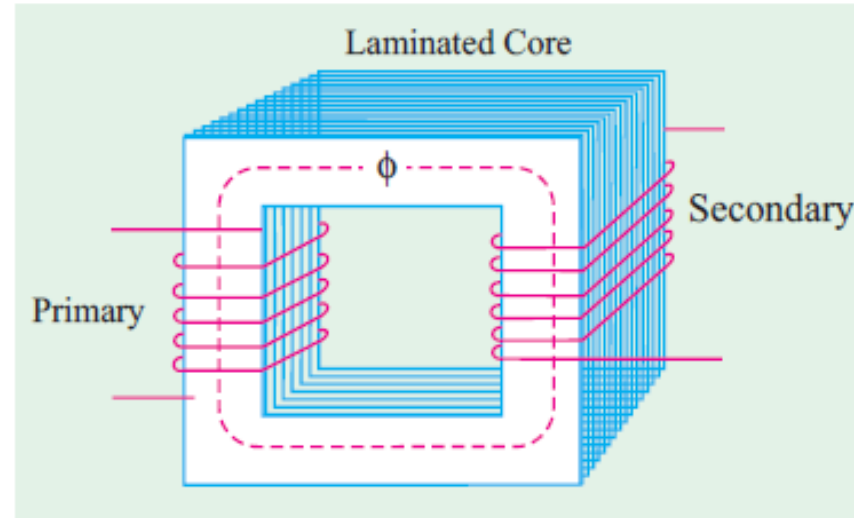


Fig. 32.1

1. transfers electric power from one circuit to another
2. it does so without a change of frequency
3. it accomplishes this by electromagnetic induction and
4. where the two electric circuits are in mutual inductive influence of each other.

32.6. E.M.F. Equation of a Transformer

Let N_1 = No. of turns in primary
 N_2 = No. of turns in secondary
 Φ_m = Maximum flux in core in webers
 $= B_m \times A$
 f = Frequency of a.c. input in Hz

As shown in Fig. 32.14, flux increases from its zero value to maximum value Φ_m in one quarter of the cycle *i.e.* in $1/4f$ second.

$$\begin{aligned}\therefore \text{Average rate of change of flux} &= \frac{\Phi_m}{1/4f} \\ &= 4f\Phi_m \text{ Wb/s or volt}\end{aligned}$$

Now, rate of change of flux per turn means induced e.m.f. in volts.

$$\therefore \text{Average e.m.f./turn} = 4f\Phi_m \text{ volt}$$

If flux Φ varies *sinusoidally*, then r.m.s. value of induced e.m.f. is obtained by multiplying the average value with form factor.

$$\text{Form factor} = \frac{\text{r.m.s. value}}{\text{average value}} = 1.11$$

$$\therefore \text{r.m.s. value of e.m.f./turn} = 1.11 \times 4f\Phi_m = 4.44f\Phi_m \text{ volt}$$

Now, r.m.s. value of the induced e.m.f. in the whole of primary winding

$$= (\text{induced e.m.f./turn}) \times \text{No. of primary turns}$$

$$E_1 = 4.44fN_1\Phi_m = 4.44fN_1B_mA$$

...(i)

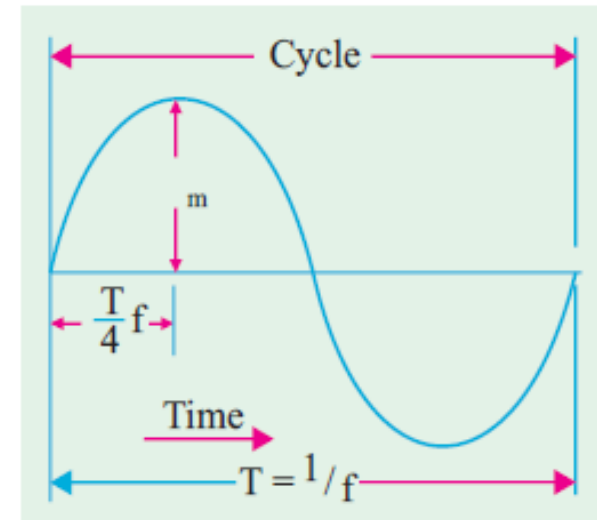


Fig. 32.14

32.7 Voltage Transformation Ratio (K)

From equations (i) and (ii), we get

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

This constant K is known as voltage transformation ratio.

(i) If $N_2 > N_1$ i.e. $K > 1$, then transformer is called *step-up* transformer.

(ii) If $N_2 < N_1$ i.e. $K < 1$, then transformer is known as *step-down* transformer.

Again, for an *ideal* transformer, input VA = output VA .

$$V_1 I_1 = V_2 I_2 \text{ or } \frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{1}{K}$$

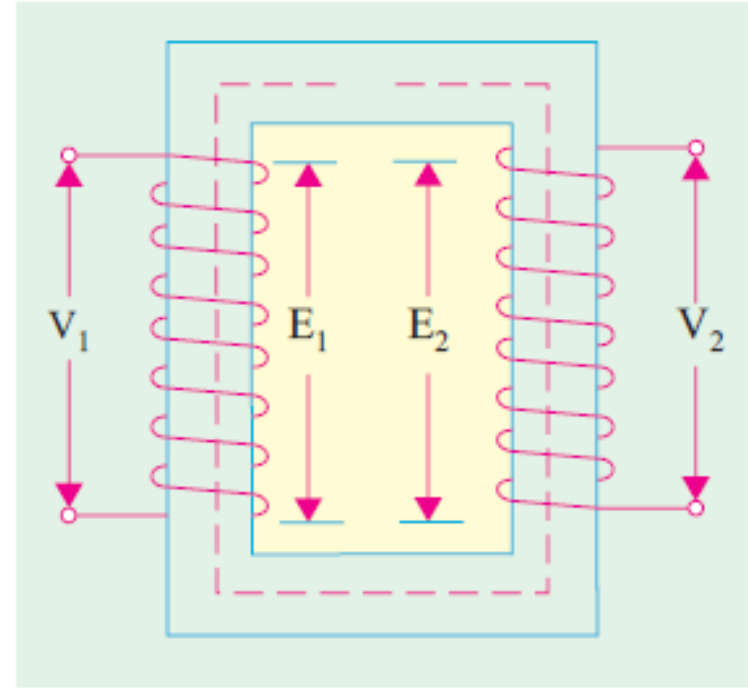


Fig. 32.15

Example 32.3. A single-phase transformer has 400 primary and 1000 secondary turns. The net cross-sectional area of the core is 60 cm^2 . If the primary winding be connected to a 50-Hz supply at 520 V, calculate (i) the peak value of flux density in the core (ii) the voltage induced in the secondary winding.

Solution.

$$K = N_2/N_1 = 1000/400 = 2.5$$

(i) $E_2/E_1 = K \therefore E_2 = KE_1 = 2.5 \times 520 = \mathbf{1300 \text{ V}}$

(ii) $E_1 = 4.44 f N_1 B_m A$

or $520 = 4.44 \times 50 \times 400 \times B_m \times (60 \times 10^{-4}) \therefore B_m = \mathbf{0.976 \text{ Wb/m}^2}$

Example 32.4. A 25-kVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000-V, 50-Hz supply. Find the full-load primary and secondary currents, the secondary e.m.f. and the maximum flux in the core. Neglect leakage drops and no-load primary current.

Solution.

$$K = N_2/N_1 = 50/500 = 1/10$$

Now, full-load $I_1 = 25,000/3000 = \mathbf{8.33 \text{ A}}$. F.L. $I_2 = I_1/K = 10 \times 8.33 = \mathbf{83.3 \text{ A}}$

e.m.f. per turn on primary side $= 3000/500 = 6 \text{ V}$

$\therefore$ secondary e.m.f. $= 6 \times 50 = \mathbf{300 \text{ V}}$ (or $E_2 = KE_1 = 3000 \times 1/10 = 300 \text{ V}$)

Also, $E_1 = 4.44 f N_1 \Phi_m$; $3000 = 4.44 \times 50 \times 500 \times \Phi_m \therefore \Phi_m = \mathbf{27 \text{ mWb}}$